UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 06/06/2019*

EXERCISE

We observe N samples of the following discrete-time real random process:

$$X[n] = S[n] + W[n] = A\left[1+\cos\left(2\pi f_0 n+\theta\right)\right] + W[n], \quad n = 0,1,\cdots,N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a Gaussian distributed random parameter with mean 0 and variance σ_A^2, the initial phase θ is modelled as a random parameter uniformly distributed in $[-\pi,\pi)$, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_W\left(e^{j2\pi f}\right) = \sigma_W^2$. The amplitude A, the phase θ, and the noise samples are mutually independent. The frequency f_0 is an a-priori known deterministic parameter.

(1) Derive the Autocorrelation Function (ACF) and the Power Spectral Density (PSD) of the w.s.s. process $X[n]$ and plot them.

(2) Derive the Linear Minimum Mean Square Error (LMMSE) estimator of the parameter A.

(3) Derive the bias and the Mean Square Error (MSE) of the LMMSE estimator of A and express it as a function of $\gamma \triangleq \dfrac{\sigma_A^2}{\sigma_W^2}$.

(4) Calculate in closed form the estimator and its MSE numerically when N=2, $\sigma_A^2 = 1$, $\gamma \triangleq \dfrac{\sigma_A^2}{\sigma_W^2} = 2$ and $f_0 = \dfrac{1}{6}$.

Solution

(1) ACF:

$$R_X[m] = R_S[m] + R_W[m] = R_S[m] + \sigma_W^2 \delta[m]$$

$$\begin{aligned} R_S[m] &= E\left\{A^2\left[1+\cos\left(2\pi f_0 n + \theta\right)\right]\left[1+\cos\left(2\pi f_0 (n+m)+\theta\right)\right]\right\} \\ &= E\left\{A^2\right\} E\left\{\left[1+\cos\left(2\pi f_0 n + \theta\right)\right]\left[1+\cos\left(2\pi f_0 (n+m)+\theta\right)\right]\right\} \\ &= \sigma_A^2\left[1+\frac{1}{2}\cos\left(2\pi f_0 m\right)\right] = \sigma_A^2 + \frac{\sigma_A^2}{2}\cos\left(2\pi f_0 m\right) \end{aligned}$$

$$R_X[m] = \sigma_A^2 + \frac{\sigma_A^2}{2}\cos\left(2\pi f_0 m\right) + \sigma_W^2 \delta[m]$$

PSD:

$$\begin{aligned} S_X\left(e^{j2\pi f}\right) &= FT\left\{R_X[m]\right\} \\ &= 2\pi\sigma_A^2 \delta\left(e^{j2\pi f} - 1\right) + 2\pi\frac{\sigma_A^2}{4}\delta\left(e^{j2\pi f} - e^{j2\pi f_0}\right) + 2\pi\frac{\sigma_A^2}{4}\delta\left(e^{j2\pi f} - e^{-j2\pi f_0}\right) + \sigma_W^2 \end{aligned}$$

(2) LMMSE estimator of A.

$$\hat{A}_{LMMSE} = \mathbf{a}^T\mathbf{X} + b\,, \quad E\{A\} = 0, \quad E\{\mathbf{X}\} = \mathbf{0} \quad \rightarrow \quad b = 0$$

$$\hat{A}_{LMMSE} = \mathbf{a}^T\mathbf{X} \quad \text{where} \quad \mathbf{a} = \mathbf{C}_X^{-1}\mathbf{c}_{XA}$$

$$\mathbf{C}_X = \mathbf{C}_S + \mathbf{C}_W = \mathbf{C}_S + \sigma_W^2\mathbf{I}, \quad \text{where} \quad \left[\mathbf{C}_S\right]_{n.n+m} = R_S[m] = \sigma_A^2 + \frac{\sigma_A^2}{2}\cos\left(2\pi f_0 m\right)$$

$$\mathbf{c}_{XA} = E\{\mathbf{X}A\} \quad \rightarrow \quad \left[\mathbf{c}_{XA}\right]_{n+1} = E\{X[n]A\} = E\left\{A^2\left[1+\cos\left(2\pi f_0 n+\theta\right)\right] + W[n]A\right\} = \sigma_A^2$$

$$\mathbf{c}_{XA} = \sigma_A^2\mathbf{1}$$

$$\hat{A}_{LMMSE} = \sigma_A^2\mathbf{1}^T\left(\mathbf{C}_S + \sigma_W^2\mathbf{I}\right)^{-1}\mathbf{X} = \mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1}\mathbf{X}$$

where: $\gamma \triangleq \dfrac{\sigma_A^2}{\sigma_W^2}$ and $\left[\dfrac{1}{\sigma_A^2}\mathbf{C}_S\right]_{n,n+m} = 1 + \dfrac{1}{2}\cos(2\pi f_0 m)$

(3) Bias and MSE of the LMMSE estimator of A.

$$E\{A\} = 0, \quad E\left\{\hat{A}_{LMMSE}\right\} = \mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1} E\{\mathbf{X}\} = 0 \quad \rightarrow \quad \hat{A}_{LMMSE} \text{ is unbiased}$$

$$MMSE\left\{\hat{A}_{LMMSE}\right\} = E\left\{\left(A - \hat{A}_{LMMSE}\right)^2\right\} = E\left\{A\left(A - \hat{A}_{LMMSE}\right)\right\} = E\left\{A^2\right\} - E\left\{A\hat{A}_{LMMSE}\right\}$$

$$E\left\{A\hat{A}_{LMMSE}\right\} = \mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1} E\{\mathbf{X}A\} = \mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1} \mathbf{c}_{AX} = \sigma_A^2\mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1}\mathbf{1}$$

$$MMSE\left\{\hat{A}_{LMMSE}\right\} = \sigma_A^2 - \sigma_A^2\mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1}\mathbf{1} = \sigma_A^2\left[1 - \mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1}\mathbf{1}\right]$$

(4) LMMSE estimator of A, bias and MSE when N=2, $\sigma_A^2 = 1$, $\gamma \triangleq \dfrac{\sigma_A^2}{\sigma_W^2} = 2$ and $f_0 = \dfrac{1}{6}$.

$$\hat{A}_{LMMSE} = \mathbf{1}^T\left(\frac{1}{\sigma_A^2}\mathbf{C}_S + \frac{1}{\gamma}\mathbf{I}\right)^{-1}\mathbf{X} = [1 \quad 1]\begin{bmatrix} \frac{3}{2} + \frac{1}{\gamma} & 1 + \frac{1}{2}\cos(2\pi f_0) \\ 1 + \frac{1}{2}\cos(2\pi f_0) & \frac{3}{2} + \frac{1}{\gamma} \end{bmatrix}^{-1} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix}$$

$$= [1 \quad 1]\begin{bmatrix} \frac{3}{2} + \frac{1}{2} & 1 + \frac{1}{4} \\ 1 + \frac{1}{4} & \frac{3}{2} + \frac{1}{2} \end{bmatrix}^{-1} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = [1 \quad 1]\begin{bmatrix} 2 & \frac{5}{4} \\ \frac{5}{4} & 2 \end{bmatrix}^{-1} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix}$$

$$= \frac{1}{2}[1 \quad 1]\begin{bmatrix} 1 & \frac{5}{8} \\ \frac{5}{8} & 1 \end{bmatrix}^{-1} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \frac{1}{2}\cdot\frac{1}{1 - \frac{25}{64}}[1 \quad 1]\begin{bmatrix} 1 & -\frac{5}{8} \\ -\frac{5}{8} & 1 \end{bmatrix}\begin{bmatrix} X[0] \\ X[1] \end{bmatrix}$$

$$\hat{A}_{LMMSE} = \frac{1}{2} \cdot \frac{1}{1-\frac{25}{64}} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{5}{8} \\ -\frac{5}{8} & 1 \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \frac{32}{39} \begin{bmatrix} \frac{3}{8} & \frac{3}{8} \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix}$$

$$= \frac{32}{39} \cdot \frac{3}{8} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \frac{4}{13} \left(X[0] + X[1] \right)$$

$$= \frac{8}{13} \cdot \frac{X[0] + X[1]}{2} = 0.3077 \cdot \left(X[0] + X[1] \right)$$

$$MMSE\left\{ \hat{A}_{LMMSE} \right\} = \sigma_A^2 \left[1 - \mathbf{1}^T \left(\frac{1}{\sigma_A^2} \mathbf{C}_S + \frac{1}{\gamma} \mathbf{I} \right)^{-1} \mathbf{1} \right]$$

$$= 1 - \frac{4}{13} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 - \frac{8}{13} = \frac{5}{13} = 0.3846$$